



BUSINESS ANALYSIS

Lecture 2: Introduction to Some Important Business Frameworks

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RATIONALE FOR BUSINESS METHODOLOGIES/Frameworks

- General principles all businesses should focus on to optimize profits, output and service quality:
 - Process efficiency
 - Reducing defects
 - Maximizing customer/client satisfaction
- Business methodologies/philosophies/frameworks aim at enshrining these principles
- The two-way relationship between business analytics and business methodologies
 - Methodologies provide the framework for how business work is organized
 - Analytics provides insights needed to make the methodologies effective
- Examples of the relationship between business principles and business analytics:
 - Manufacturing expenses and profitability
 - Process efficiency vis-à-vis customer satisfaction

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BUSINESS FRAMEWORKS FOR OPTIMIZING PROFITS & OUTPUT/SERVICE QUALITY

- **Total Quality Management**
A customer-focused approach that aims to build quality in from the start, reduce errors, and enhance efficiency
- **Six Sigma**
A data-driven method to improve process quality by reducing defects
- **ISO 9000**
A framework for a Quality Management System (QMS), built on 7 quality management principles like customer focus, leadership, process approach, and continuous improvement
- **Lean**
Aims to maximize customer value by eliminating waste (time, resources, defects)
- **Kanban**
Visualizes workflow (To Do, Doing, Done) to manage flow and limit work-in-progress, emphasizing continuous delivery
- **Waterfall**
A linear approach where each phase (design, development, testing) must be completed before the next begins
- **DevOps**
Combines software development (Dev) and IT operations (Ops) for faster, more reliable software delivery through automation and collaboration

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TOTAL QUALITY MANAGEMENT: HISTORY

- **Early Roots (1920s-1940s)**
Walter Shewhart (1924): Applied statistics to quality control at Bell Telephone, introducing control charts to understand process variation, laying the foundation for Statistical Quality Control (SQC)
- **The Japanese Transformation (1950s-1960s) after WWII**
 - Deming & Juran: Invited to Japan after WWII, Deming taught statistical methods; Juran focused on managerial issues, helping Japanese companies rebuild with a focus on quality
 - Armand Feigenbaum: Published *Total Quality Control*, broadening quality beyond engineering to all functions
 - Kaoru Ishikawa: Developed *Company-Wide Quality Control* (CWQC) in Japan

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TOTAL QUALITY MANAGEMENT: PHILOSOPHICAL CONTRIBUTION

- Key Figures
 - W. Edwards Deming: Statistical control, focus on systems and management
 - Joseph M. Juran: Quality planning, control, improvement, and management
 - Armand V. Feigenbaum: Total Quality Control concept
 - Kaoru Ishikawa: Quality circles, contribution to Japanese Company Wide Quality Control (CWQC)
 - Philip Crosby: "Zero Defects" philosophy
- Modern Influence

TQM principles are now foundational to modern quality management systems (like ISO 9000) and business excellence models, emphasizing continuous improvement (Kaizen) and customer satisfaction

Deming and Post-WWII Japanese Reconstruction: <https://www.youtube.com/watch?v=7nw8dydDvgA>

Japan and the Quality Revolution by Joseph M. Juran: <https://www.youtube.com/watch?v=28YuuljDuto>

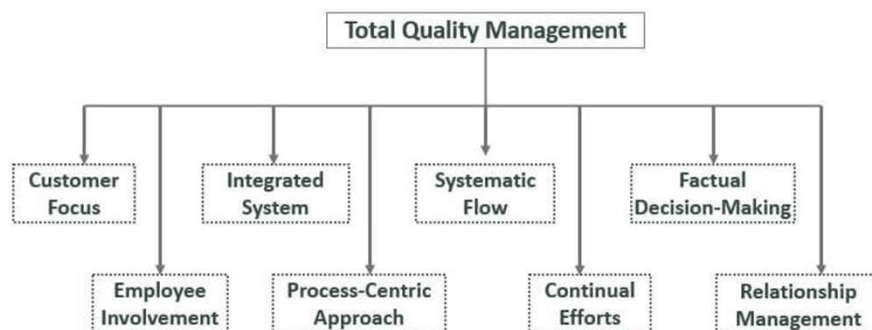
Steve Jobs on Joseph Juran and Quality: <https://www.youtube.com/watch?v=XbkMcvnNq3g>

Tokyo the 1945 Fire Bombing: Then and Now: <https://apimagesblog.com/blog/2015/03/10/tokyo-1945-firebombing-then-and-now>

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8 PRINCIPLES OF TQM

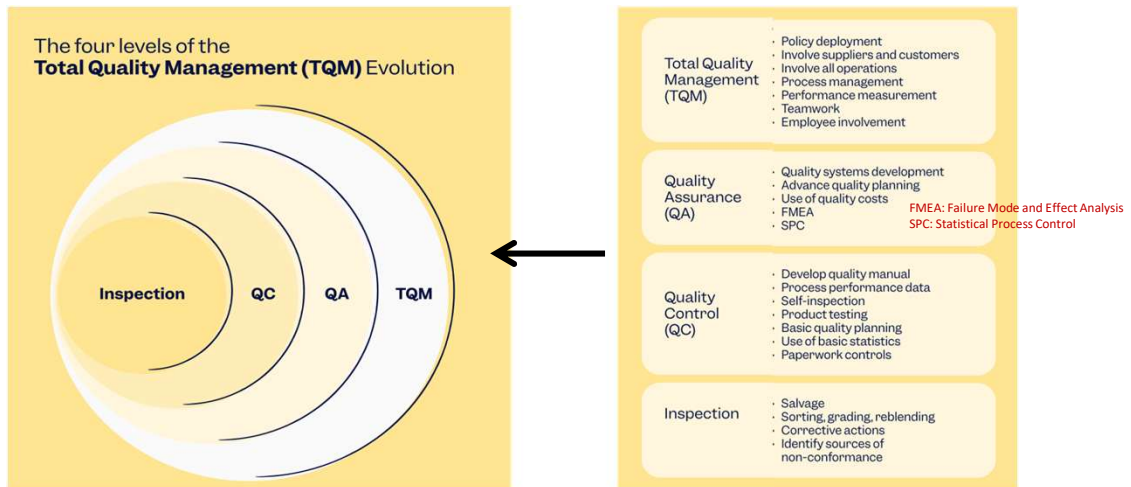
Principles of Total Quality Management



Source: WallStreetMojo; Available at: <https://www.wallstreetmojo.com/total-quality-management/>

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IMPLEMENTING TQM IN AN ORGANIZATION



Source: scilife; Available at: <https://www.scilife.io/blog/total-quality-management>

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SIX SIGMA: HISTORY

- “Introduced by Bill Smith at Motorola in 1986 to improve manufacturing quality
- Motorola registered it as a trademark in the early 1990s
- Companies like Honeywell and General Electric (GE) adopted it, with GE’s CEO Jack Welch making it central to his strategy in 1995
- By the late 1990s, about two-thirds of Fortune 500 companies had started Six Sigma initiatives”

Source: SIX SIGMA ONLINE; available at: <https://www.sixsigmaonline.org/six-sigma-history/>

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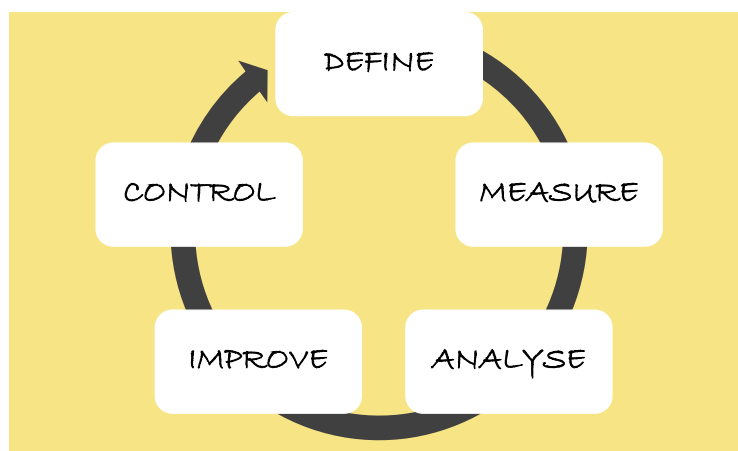
EXPLAINING SIX SIGMA

- What is Six Sigma?
 “Six Sigma is a data-driven methodology that aims to improve processes by identifying and eliminating defects or variations, ultimately striving for near-perfect quality with only 3.4 defects per million opportunities”
- Benefits of Six Sigma
 - Elimination of expenses associated with errors and defects
 - Increase in the quality of products or services
 - Increase in profitability
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 - Elimination of expenses associated with errors and defects
 - Increase in the quality of products or services
 - Increase in profitability

Source: SIX SIGMA ONLINE; available at: <https://www.sixsigmaonline.org/six-sigma-history/>

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IMPLEMENTING SIX SIGMA IN AN ORGANIZATION



Source: scilife; Available at: <https://www.scilife.io/blog/total-quality-management>

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INTERNATIONAL STANDARDIZATION ORGANIZATION (ISO): HISTORY

- History of ISO
 - 1946: London, 65 delegates from 25 countries met to discuss the future of International Standardization.
 - 1947: ISO was officially created with 67 technical committees (groups of experts focusing on a specific subject)
- What is ISO?
 - “ISO is an independent, non-governmental international organization. It brings global experts together to agree on the best ways of doing things”
- Examples of Standards
 - ISO 9001: Quality Management Systems.
 - ISO 27001: Information Security Management Systems.
 - ISO 14001: Environmental Management Systems.
 - ISO 37001: Anti-bribery Management Systems

Source: ISO; available at: <https://www.iso.org/home.html>

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ISO 9001: QUALITY MANAGEMENT SYSTEMS

- What is ISO 9001?
 - “ISO 9001 is a globally recognized standard for quality management. It helps organizations of all sizes and sectors to improve their performance, meet customer expectations and demonstrate their commitment to quality. Its requirements define how to establish, implement, maintain, and continually improve a quality management system (QMS).”
- Benefits of ISO 9001 Standardization
 - “Implementing ISO 9001 means your organization has put in place effective processes and trained staff to deliver flawless products or services time after time.”
 - Increased customer trust and satisfaction
 - Sound quality control processes
 - Cost savings and productivity gains
 - Culture of continuous improvement

Source: ISO; available at: <https://www.iso.org/standard/62085.html>

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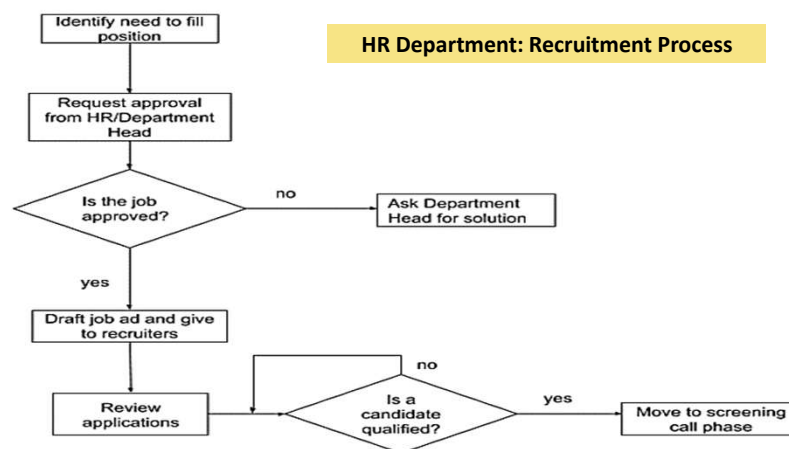
HOW DOES STANDARDIZATION WORK: ISO CERTIFICATION REQUIREMENTS

- **Prepare and Plan:**
 - Learn the Standard: Understand ISO 9001:2015 requirements and principles (e.g. customer focus, leadership, etc.)
 - Gap Analysis: Assess your current processes against ISO 9001 to identify gaps
- **Implement**
 - Define Processes: Map an organization's processes, e.g., manufacturing steps
 - Document Procedures: Create necessary documents like the Quality Policy Standard Operating Procedures (SOPs)
- **Internal Audit and Review**
 - Perform Internal Audits and Reviews: Conduct audits to check if the QMS meets standards and is effective.
 - Corrective Actions: Implement actions for nonconformities found during internal audits
- **Certification Audit**
 - Select a Body: Choose an accredited, third-party certification party (e.g., SGS (Société Générale de Surveillance) Pakistan).
 - Stage 1 Audit: The body reviews your documentation and readiness
 - Stage 2 Audit: The body audits your QMS implementation on-site, checking processes

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EXAMPLE OF DOCUMENTING A PROCESS FOR ISO CERTIFICATION



Source: Resources for Employers; Available at: <https://resources.workable.com/tutorial/faq-hiring-process>

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TEN MINUTES ACTIVITY

1. Many organizations want to pursue a goal of excellence. What would an organization need to do to become truly excellent? Identify five things you think an organization would need to do/ have in place.
2. What does the term quality management mean to you? Note at least five components or terms that you associate with quality management?
3. The previous slide demonstrated the HR recruitment process of a company. How can you improve that process? You will get a star you can draw an alternate flowchart to depict the changes.



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SWOT ANALYSIS

Strengths, Weaknesses, Opportunities, and Threats (SWOT) analysis

"A SWOT Analysis is a managerial decision making tool used to identify a firm's internal **strengths** and **weaknesses**, as well as external **threats** and **opportunities**."

Strengths

Areas that the company is performing well in or is good at such as having a strong brand image or a good corporate culture

Weaknesses

Areas that the company is not doing well in or is performing more poorly such as lack of investment in new technology or a poorly performing product.

Opportunities

Things that could happen outwith the business to help them grow or become more profitable such as the chance to take over a competitor or a boom in the economy.

Threats

External factors that could prevent a business from meeting its goals such as:

- a new competitor opening or reducing their prices
- a recession"

Source: BBC, *Decision making*; available at <https://www.bbc.co.uk/bitesize/guides/zkdc7nb/revision/5>

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SWOT ANALYSIS: THE FOUR QUADRANTS

Example from a construction industry

strengths - Highly efficient staff - High level of customer satisfaction - No. of contracts have grown from PKR 30 million to PKR 150 million	weaknesses - New technology not being adopted - Limited capacity to take on new contracts - Weak marketing
S	W
O - Property market is booming - Import of cheap construction material growing easy - Government is supporting investment in low-cost housing	T - Labor costs are rising - Local supply chain of cement is facing disruptions - Bank interest rates are rising
opportunities	threats

Source; adapted from: smartsheet; available at https://www.smartsheet.com/content/how-to-do-swot-analysis?srsltid=AfmBOopN_I_ku4q4pncsH-MZZAJcY1QKbV5P4_E-5XcM86FQDQWh_xzQ

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SWOT ANALYSIS: 10 MINUTES ACTIVITY

HOW TO CONDUCT A PERSONAL SWOT ANALYSIS



Source: Brad Sugars; available at <https://bradsugars.com/personal-swot-analysis/>

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ESSENTIAL READINGS

WallStreetMojo

Page Title: Total Quality Management

Available at: <https://www.wallstreetmojo.com/total-quality-management/>

Scilife

Page Title: Mastering Total Quality Management in your organization

Available at: <https://www.scilife.io/blog/total-quality-management>

Links

- [Key takeaways](#)
- [Total Quality Management definition](#)
 - [Understanding Total Quality Management](#)
- [Benefits of Total Quality Management](#)
- [The four levels of the Total Quality Management evolution](#)
 - [Level 1. Inspection](#)
 - [Level 2. Quality Control](#)
 - [Level 3. Quality Assurance](#)
 - [Level 4. Total Quality Management](#)
- [The 8 Total Quality Management principles](#)

SIX SIGMA ONLINE

Page Title: History of Six Sigma: Exploring the Roots

Available at: <https://www.sixsigmaonline.org/six-sigma-history/>

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ESSENTIAL READINGS

SIXSIGMA INSTITUTE

Page Title: What is Six Sigma and Why is it Six Sigma

Available at: https://www.sixsigma-institute.org/What_Is_Sigma_And_Why_Is_It_Six_Sigma.php

ISO

Page Title: ISO: Global standards for trusted goods and services

Available at: <https://www.iso.org/home.html>

Links

- [Standards](#)
- [Sectors](#)
- [About ISO](#)
- [Insights & news](#)
- [Taking part](#)
- [Store](#)

ISO

Page Title: ISO 9001:2015(en) Quality management systems — Requirements

Available at: <https://www.iso.org/obp/ui/en/#iso:std:iso:9001:ed-5:v1:en>

BBC

Page Title: Decision making: SWOT analysis

Available at: <https://www.bbc.co.uk/bitesize/guides/zkdc7nb/revision/5>

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SUGGESTED READINGS

ISO

Page Title: ISO/IEC 42001:2023(en) Information technology — Artificial intelligence — Management system

Available at: <https://www.iso.org/obp/ui/en/#iso:std:iso-iec:42001:ed-1:v1:en>

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LECTURE 3

- Introduction to Four types of analysis
 - Descriptive
 - Diagnostic
 - Predictive
 - Prescriptive
- What is Descriptive Analysis
- What is Diagnostic Analysis

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